

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Allows users to securely log in and access the application.	Authentication Module	Done	10%
2	Audio Upload & Recording	Enables users to upload or record voice explanations.	User Interface	Done	15%
3	Speech-to-Text Conversion	Converts spoken audio into text using Whisper AI.	Speech Processing	In Progress	15%
4	Concept Understanding Analysis	Compares the transcript with the expected answer using Sentence-BERT.	AI Analysis Module	Pending	20%
5	Audio Feature Extraction	Analyzes speech characteristics such as fluency, pauses, and speaking rate.	Audio Analysis Module	Pending	10%
6	AI Feedback Generation	Generates personalized feedback and improvement suggestions using Gemini AI.	AI Feedback Module	Pending	10%
7	Results Dashboard	Displays transcript, semantic score, waveform, and evaluation results.	Dashboard Module	In Progress	10%
8	PDF Report Generation	Generates and downloads a detailed performance report.	Report Generation Module	Pending	10%

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	2
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	2

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Home Page, Audio Upload, Recording, Dashboard	Upload Audio, Results Screen
2	Backend / Logic	Speech Processing, Semantic Analysis, Scoring	Whisper Transcription, Sentence-BERT Evaluation
3	Database / Storage	User Data, Reports, Session History	SQLite/PostgreSQL Storage
4	API / Integration	AI Services & Audio Processing	Whisper API, Gemini API, Librosa
5	Security / Authentication	User Login and Session Management	Login Authentication, Secure Access